Referee B – Comment 1

> *In describing the approach used to calculate the radiative widths of axial-vector meson states, the authors refer to their previous work on estimating the mixing angle of the $^1P_1$ and $^3P_1$ spin states (Phys. Rev. D **111**, 014005 (2025)). In my view, it would be beneficial for the reader if the authors briefly outlined this approach and the main results in the present article, taking into account—however slight—the changes in the input parameter values (such as the quark condensate magnitude) used in these studies.*

Response:

We thank the referee for this helpful suggestion. We have expanded Sec. II, following the two-point sum rules for the decay constants, to summarize the mixing-angle determination in our earlier work. The added discussion, Eqs. (39)–(43) of the revised manuscript, explains the current rotation, the diagonalization condition, and the role of the physical diagonal correlators. The resulting mixing angles are quoted in Eq. (46).

We also clarify a typographical omission in the earlier article. The tensor-current normalization factor $1/(m_Q+m_s)$, with $Q = c, b$, was included in the calculations but was inadvertently omitted from the printed current definition. Thus, the two studies use the same current normalization. This clarification has also been included in a footnote in the revised manuscript.

Following the referee's suggestion, we assessed the effect of the different condensate inputs through a paired numerical comparison using a reconstruction of the earlier sum-rule expressions.

We changed the central light-quark condensate at $\mu = 1\,\mathrm{GeV}$ from

$$\langle\bar{q}q\rangle_{\mathrm{old}} = -0.0165\,\mathrm{GeV}^3 \quad \text{to} \quad \langle\bar{q}q\rangle_{\mathrm{new}} = -(0.240)^3\,\mathrm{GeV}^3. \tag{1}$$

The strange and mixed condensates were changed consistently through $\langle\bar{s}s\rangle = 0.8\langle\bar{q}q\rangle$ and $\langle\bar{s}g_s\sigma Gs\rangle = m_0^2\langle\bar{s}s\rangle$, with $m_0^2 = 0.8\,\mathrm{GeV}^2$, while all other inputs and the sum-rule prescription were held fixed.

At the working-window midpoints, $(M^2, s_0) = (4.5, 10)\,\mathrm{GeV}^2$ for D_s and $(11, 43)\,\mathrm{GeV}^2$ for B_s, the shifts $\Delta\theta = \theta_{\mathrm{new}} - \theta_{\mathrm{old}}$ are

$$\Delta\theta_{D_s} = +0.185^\circ, \qquad \Delta\theta_{B_s} = +0.057^\circ. \tag{2}$$

Within this comparison, the shifts remain smaller than the previously quoted uncertainties of 0.6° and 0.1° throughout the sampled regions. We therefore retain the published mixing angles and their quoted uncertainties as external inputs in the present analysis. This comparison tests sensitivity to the central condensate inputs; it does not constitute a new extraction of the mixing angles or a revision of their uncertainties.

Referee B – Comment 2

It seems to me that it would be very interesting to obtain—within the framework of the approach proposed by the authors—the values of the radiative decay widths into the vector D_s^* and B_s^* meson states. Of course, the calculation of these widths might well warrant a separate paper. However, the authors should, in my view, note that measuring absolute widths is a far more challenging experimental task than measuring the ratio of radiative decay widths into two different final states. This holds particularly true for the P-wave states of B_s mesons. One would also expect the ratio of decay probabilities into $D_s\gamma$ and $D_s^*\gamma$ for the $D_{s1}(2536)$ to depend strongly on the mixing angle.

Response:

We thank the referee for this useful suggestion. We agree that ratios of radiative widths into pseudoscalar and vector final states are experimentally more accessible than absolute widths, particularly for the bottom-strange mesons, and may provide complementary information on the mixing angle.

A separate study of radiative transitions into the vector states D_s^* and B_s^* is nearing completion. Following the referee's suggestion, that study will also present ratios of the radiative widths into vector and pseudoscalar final states and examine their mixing-angle dependence, with particular attention to the $D_{s1}(2536)$ channel.

Referee B – Comment 3

The decay constants f_1 and f_2 for the physical states $D_{s1}(2460)$ and $D_{s1}(2536)$, obtained in the paper as a byproduct, are of independent interest; they allow for the calculation—within the factorization approximation—of the ratio of the branching fractions for the B-meson decays into $D_{s1}(2536)D^-$ and $D_{s1}(2460)D^-$. As evidenced by LHCb results (arXiv:2411.03399), this ratio is experimentally accessible, given that the branching fractions for the decays of both $D_{s1}(2460)$ and $D_{s1}(2536)$ into $D_s\pi^+\pi^-$ have already been measured. It would be interesting to present the dependence of the ratio f_2/f_1 on the mixing angle in the paper.

Response:

We thank the referee for highlighting the independent interest of the decay constants and their potential application to nonleptonic B decays.

We have added Fig. 1 in Sec. III of the revised manuscript, showing $f_1(\theta)$ and $f_2(\theta)$ separately for both the charm-strange and bottom-strange sectors. This displays the individual quantities entering our radiative-decay sum rules. The corresponding ratio $f_2(\theta)/f_1(\theta)$ requested by the referee is presented in Fig. 1 of this response.

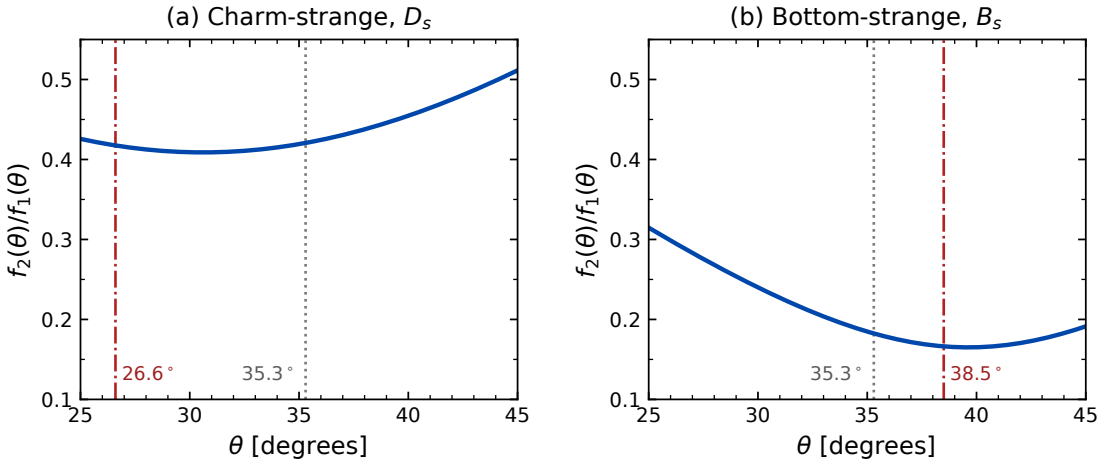


Figure 1: Mixing-angle dependence of the ratio $f_2(\theta)/f_1(\theta)$.